

# Ouranos Reconstruction – RDPS-CaPA (ORRC) v1.0 / Reconstruction Ouranos – RDPS-CAPA (RORC) v1.0

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## 1 Background and Summary

The Canadian Surface Reanalysis (CaSR) ([link](#)) dataset is a high-resolution reanalysis product developed by Environment and Climate Change Canada (ECCC) and provides an observation-based representation of surface and near-surface atmospheric conditions over North America, Central America, and nearly all of Greenland (Gasset et al., 2021; Khedhaouria et al., 2026). CaSR version 2.1 covers 1980–2018, and CaSR version 3.2 extends the record to 2024. Continuous climate monitoring and operational climate services, however, require high-resolution reanalysis data through the present day. To ensure continuity beyond the CaSR period, we produced the Ouranos Reconstruction – RDPS-CaPA (ORRC), version 1.0. The methodological framework was designed by ECCC to generate a CaSR-like dataset from the Regional Deterministic Prediction System (RDPS; [link](#)) and the Regional Deterministic Precipitation Analysis (RDPA; [link](#)), which is based on the Canadian Precipitation Analysis (CaPA) system. Ouranos implemented the end-to-end data production. ORRC v1.0 provides hourly surface and near-surface meteorological fields over the same domain as CaSR v3.2 at a resolution of 0.09° (~10 km) on a rotated uniform latitude-longitude grid, over land and ocean, from 2015 to present.

ORRC's primary contribution is temporal continuity at high spatial and temporal resolution, enabling near-real-time climate monitoring and applications that compare the most recent conditions against historical baselines. This is particularly valuable for operational climate services, event monitoring, contextual analysis, and evidence-informed planning, allowing practitioners to assess recent events – such as heatwaves, heavy rainfall, or droughts – in a long-term context without waiting for official reanalysis updates.

## 2 Methodology

### 2.1 Data

#### 2.1.1 Regional Deterministic Prediction System (RDPS)

RDPS is ECCC's operational short-range deterministic numerical weather prediction system and was formerly based on a limited-area configuration of the Global Environmental Multiscale (GEM) model with a 10-km grid spacing covering North America (ECCC, 2024b). Forecast outputs are provided on a regional rotated latitude–longitude grid covering North America and adjacent oceans at 0.09° (~10 km) horizontal resolution. The system produces forecasts of near-surface and atmospheric variables, including temperature, precipitation, cloud cover, wind speed and direction, humidity and others. Forecasts extend to 84 hours and are issued four times per day at 00, 06, 12, and 18 UTC. To construct ORRC, we retain only the 6- to 12-hour forecast lead times from each RDPS cycle to minimize spin-up errors and model drift.

#### 2.1.2 Canadian Precipitation Analysis (CaPA-RDPA)

In its deterministic configuration, CaPA produces the Regional Deterministic Precipitation Analysis (RDPA), providing optimal estimates of accumulated precipitation over 6-hour periods, valid at 00, 06, 12, and 18 UTC, and 24-hour periods, valid at 06 and 12 UTC, on a ~10 km grid over North America (Canada, United States, Mexico) (ECCC, 2024a). RDPA combines surface station observations, radar precipitation estimates, IMERG satellite precipitation estimates, and a trial field from RDPS using an optimal interpolation framework, also known as residual kriging (Fortin et al., 2018). The analysis uses innovations (difference between observations and trial field), with error statistics updated at each production run, and includes quality control procedures for the assimilated observations and precipitation estimates.

Table 1 places RDPA within the broader set of CaPA configurations. CaPA is used in both operational near-real-time products and reanalysis-based products, with differences in background field, spatial resolution, accumulation period, domain, temporal availability, and assimilated data sources. In ORRC production, the operational CaPA-RDPA 6- and 24-hour precipitation analyses are used as reference accumulations to adjust RDPS precipitation and produce a CaSR-like precipitation field. For simplicity, the RDPA 6- and 24-hour analyses used in ORRC production are referred to as CaPA-6h and CaPA-24h, respectively, throughout this document.

Table 1. Comparison of the operational CaPA-RDPA product and the online and offline CaPA components used in the CaSR reanalysis.

|                             | Operational CaPA       | CaPA in CaSR reanalysis            |                                   |
|-----------------------------|------------------------|------------------------------------|-----------------------------------|
|                             | RDPA                   | Online component (surface coupled) | Offline component (posteriori)    |
| <b>Configuration</b>        | Deterministic          | Deterministic                      | Deterministic                     |
| <b>Spatial resolution</b>   | 10 km                  | 10 km                              | 10 km                             |
| <b>Accumulation period</b>  | 6 h / 24 h             | 6 h                                | 24 h; hourly disaggregated        |
| <b>Spatial coverage</b>     | North America          | North America                      | North America                     |
| <b>Temporal coverage</b>    | 2011–present           | 1968–2024                          | 1968–2024                         |
| <b>Surface station data</b> | Yes                    | Yes                                | Yes + additional climate networks |
| <b>Radar data</b>           | Yes (warm season only) | No                                 | No                                |
| <b>Satellite data</b>       | Yes (warm season only) | No                                 | No                                |

### 2.1.3 Data retrieval

CaPA-24h data for the full period, along with CaSR v3.2 data for a single date, were downloaded from ECCC's high-performance computer GPSC-C and converted to NetCDF format<sup>1</sup>. RDPS and CaPA-6h data up to March 31<sup>st</sup>, 2025 were downloaded in NetCDF format from the Canadian Surface Prediction Archive (CaSPAr) at <https://caspar-data.ca> (Mai et al., 2020) under the ECCC Data Servers End-use Licence ([link](#)). Because of CaSPAr service unavailability, RDPS and CaPA-6h data starting from April 1<sup>st</sup>, 2025 onward were downloaded from GPSC-C and converted to NetCDF.

## 2.2 Preprocessing

Prior to ORRC production, we perform a completeness and consistency check of the raw RDPS and CaPA data. For RDPS, we verify that each forecast cycle includes the seven hourly lead times (+6 to +12 h). For CaPA, we verify that 6- and 24-hour accumulation periods are

<sup>1</sup> Note that the Ouranos ORRC v1.0 data treatment workflow retrieves RDPS and CaPA data in ECCC's FST file format via a GPSC-C collaborator account. Conversion to NetCDF is achieved using the fst2nc Python utilities: <https://github.com/neishm/fstd2nc>

represented by a file containing a single analysis time step. Missing files are generated and populated with NaN values, and any missing variables in existing files are similarly filled with NaNs.

We regrid the RDPS and CaPA data to the CaSR v3.2 target grid (0.09° resolution) using nearest-neighbour interpolation. Although the input and target grids both have the ~10 km horizontal resolution, variations in grid definitions, including changes to the RDPS operational grid over time, necessitate this regridding step. Nearest-neighbour interpolation is used because it preserves original values and can accommodate varying grid structures.

## 2.3 ORRC Production

The general methodology for ORRC dataset construction follows that described by ECCC researchers for the construction of [CaSR-like files based on real-time operational products](#) (Dimetrijevic, 2023). The details for the individual steps are described in the following subsections.

### 2.3.1 Temporal assembly and variables

Each RDPS forecast cycle provides hourly fields for lead times +6 to +12. ORRC v1.0 is constructed by concatenating these selected segments from consecutive forecast cycles to form a continuous hourly time series. The resulting 24-hour reporting window spans from 13 UTC to 12 UTC of the following calendar day.

ORRC v1.0 provides an hourly record of near-surface meteorological fields, including near-surface air temperature, dew point temperature, relative humidity, specific humidity, surface air pressure, sea level pressure, wind speed and direction, downwelling shortwave and longwave radiation at the surface, and precipitation (model precipitation and precipitation adjusted with CaPA-6h and CaPA-24h totals).

The temporal coverage of ORRC v1.0 (2015–present) is determined by the availability of the underlying RDPS and CaPA inputs in the CaSPAr archives for the variables used here. The eastward and northward near-surface wind components contain erroneous values in the CaSPAr archive from June 11, 2024 onward, which seems due to a change in grid definition. Accordingly, these variables are set to NaN in ORRC v1.0 from June 11, 2024 until access to GPSC-C became available on April 1, 2025. Users requiring near-surface wind components during this interval should rely on CaSR v3.2 (1980–2024) or alternative wind products.

We additionally provide daily aggregates, including daily means and daily minima/maxima for near-surface air temperature, dew point temperature, and relative humidity. Daily values are computed from the hourly record by resampling to a daily time step using standard calendar-day bins defined by the timestamps (midnight-to-midnight).

Table 2 shows the list of variables provided in ORRC v1.0, their corresponding ECCC identifiers and output frequencies.

*Table 2. Variables provided in ORRC v1.0. The table lists the long name, ORRC variable name, corresponding ECCC variable identifier, and output frequency. Variables marked with an asterisk are also available in the bias-adjusted ORRC-a product (daily only).*

| Long name                                               | Variable name (ORRC) | Variable name (RDPS)               | Frequency                   |
|---------------------------------------------------------|----------------------|------------------------------------|-----------------------------|
| <b>Precipitation (with CaPA blending)*</b>              | pr                   | A_PR_SFC                           | Hourly + daily mean         |
| <b>Precipitation (forecast)</b>                         | prmod                | P_PR_SFC                           | Hourly + daily              |
| <b>1.5 metre temperature*</b>                           | tas                  | P_TT_10000<br>(P_TT_1.5m in CaSR)  | Hourly + daily mean/min/max |
| <b>1.5 metre dewpoint temperature</b>                   | tdps                 | P_TD_10000<br>(P_TD_1.5m in CaSR)  | Hourly + daily mean/min/max |
| <b>Near-Surface Relative Humidity (1.5m)</b>            | hurs                 | P_HR_10000<br>(P_HR_1.5m in CaSR)  | Hourly + daily mean/min/max |
| <b>Near-Surface Specific Humidity (1.5m)</b>            | huss                 | P_HU_10000<br>(P_HU_1.5m in CaSR)  | Hourly + daily mean         |
| <b>Surface Downwelling Shortwave Radiation</b>          | rsds                 | P_FB_SFC                           | Hourly + daily mean         |
| <b>Surface Downwelling Longwave Radiation</b>           | rlsds                | P_FI_SFC                           | Hourly + daily mean         |
| <b>Surface Air Pressure</b>                             | ps                   | P_P0_SFC                           | Hourly + daily mean         |
| <b>Sea Level Pressure</b>                               | psl                  | P_PN_SFC                           | Hourly + daily mean         |
| <b>Eastward Near-Surface Wind (10m)</b>                 | uas                  | P_UUC_10000<br>(P_UUC_10m in CaSR) | Hourly + daily mean         |
| <b>Northward Near-Surface Wind (10m)</b>                | vas                  | P_VVC_10000<br>(P_VVC_10m in CaSR) | Hourly + daily mean         |
| <b>Near-Surface Wind Speed (10m)</b>                    | sfcWind              | P_UVC_10000<br>(P_UVC_10m in CaSR) | Hourly + daily mean         |
| <b>Near-Surface Meteorological Wind Direction (10m)</b> | winddir              | P_WDC_10000<br>(P_WDC_10m in CaSR) | Hourly + daily mean         |

### 2.3.2 Precipitation blending

To produce a CaSR-like precipitation field, RDPS hourly precipitation is adjusted using CaPA accumulations in two steps. First, cumulative RDPS precipitation is converted to hourly increments by differencing consecutive forecast times. Within each 6-hour window, the RDPS hourly increments are scaled so that their sum matches the corresponding CaPA-6h accumulation. If the RDPS 6-hour total is missing or near zero (less than  $1 \times 10^{-6}$  m), the CaPA-6h total is instead distributed uniformly across the six hours.

The resulting hourly fields are then concatenated into a 24-hour sequence and adjusted a second time using the CaPA-24h accumulation. When the 24-hour sum is valid, all hourly values are scaled so that the final daily total matches CaPA-24h. If the 24-hour sum is missing or near zero, the CaPA-24h total is distributed uniformly across the 24 hours. This two-step procedure ensures consistency with both the 6-hour and 24-hour CaPA accumulations while preserving the RDPS hourly precipitation distribution.

Figure 1 shows an example daily precipitation field from ORRC v1.0 over the full domain, illustrating the spatial structure resolved at  $0.09^\circ$  resolution.

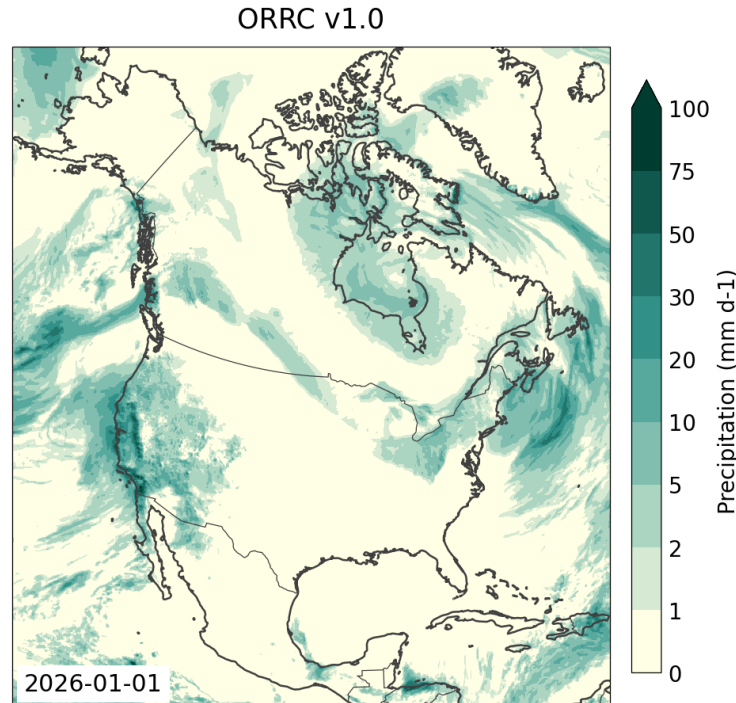


Figure 1. Daily accumulated precipitation ( $\text{mm day}^{-1}$ ) from ORRC v1.0 over North America, Central America, and nearly all of Greenland on 1 January 2026. The map illustrates the spatial structure at  $0.09^\circ$  ( $\sim 10$  km) resolution on the rotated latitude-longitude grid.

### 2.3.3 Bias adjustment

To reduce discrepancies relative to the reanalysis data, we produce a bias-adjusted variant of ORRC (ORRC-a) for a subset of daily variables (daily mean precipitation and daily mean/min/max near-surface air temperature) using CaSR v3.2 as a target dataset. For applications requiring closer consistency with CaSR v3.2, we recommend using ORRC-a v1.0 for the variables provided. The same adjustment approach can be applied to other variables if needed. Our main objective of producing bias adjusted fields is to align the near-real-time climate conditions with the historical conditions provided by CaSR, ensuring that year-to-date values are comparable and placed as accurately as possible within the long-term historical context. For applications not requiring comparison within a historical context, the unadjusted ORRC data may be suitable for direct use.

The bias adjustment is implemented using the xsdba Python library (Bourgault et al., 2026). We apply a day-of-year mean bias correction computed using a 31-day moving window ( $\pm 15$  days). For temperature variables, we apply an additive correction, and for precipitation we apply a multiplicative correction, which maintains non-negative precipitation values and preserves temporal variability around the adjusted mean. Figure 2 illustrates the impact of the bias adjustment on the monthly mean precipitation time series in a representative region over the overlapping period with CaSR v3.2.

The adjustment period is 2020–2024. The adjustment start year is selected to reflect the major upgrade of RDPS to version 7.0.0 in July 2019, which includes a change from version 4.8 to version 5.0 of the GEM atmospheric model (ECCC, 2019). This configuration is therefore the GEM version 5.2.1 employed in CaSR v3.2 (Khedhaouria et al., 2026). Starting the adjustment period in 2020 also ensures that the 31-day moving windows used to compute the day-of-year corrections are based on a consistent number of years throughout the annual cycle.

Figure 3 shows the seasonal mean precipitation bias relative to CaSR v3.2 over the full domain for ORRC v1.0 and ORRC-a v1.0 during the 2020–2024 overlap period. The bias adjustment substantially reduces the magnitude and spatial extent of precipitation biases in all seasons. Additional maps showing the effect of the bias adjustment for near-surface air temperature variables are available in the ORRC GitHub repository at <https://github.com/Ouranosinc/ORRC>.

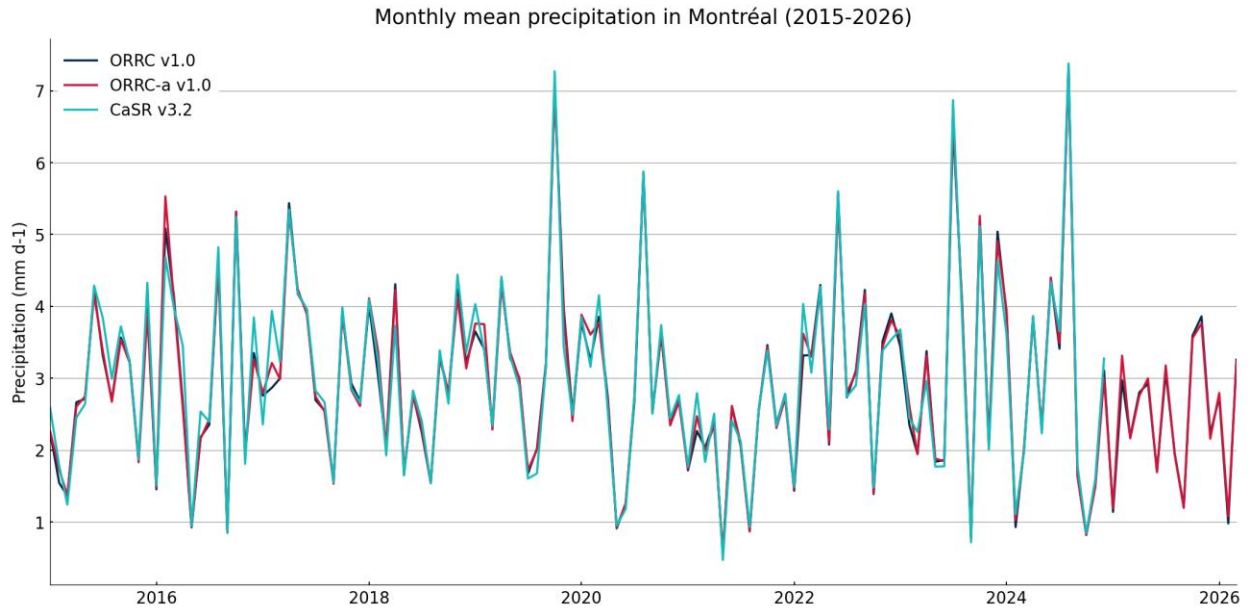


Figure 2. Monthly mean precipitation ( $\text{mm day}^{-1}$ ) in the Montréal regional county municipality for 2015–2026 derived from ORRC v1.0, ORRC-a v1.0, and CaSR v3.2. The bias-adjusted product (ORRC-a v1.0) is designed to reduce mean bias relative to CaSR v3.2 while preserving temporal variability around the mean. CaSR v3.2 is available through 2024, and ORRC v1.0 and ORRC-a v1.0 extend to the present.



### Seasonal Bias in Precipitation (2020-2024)

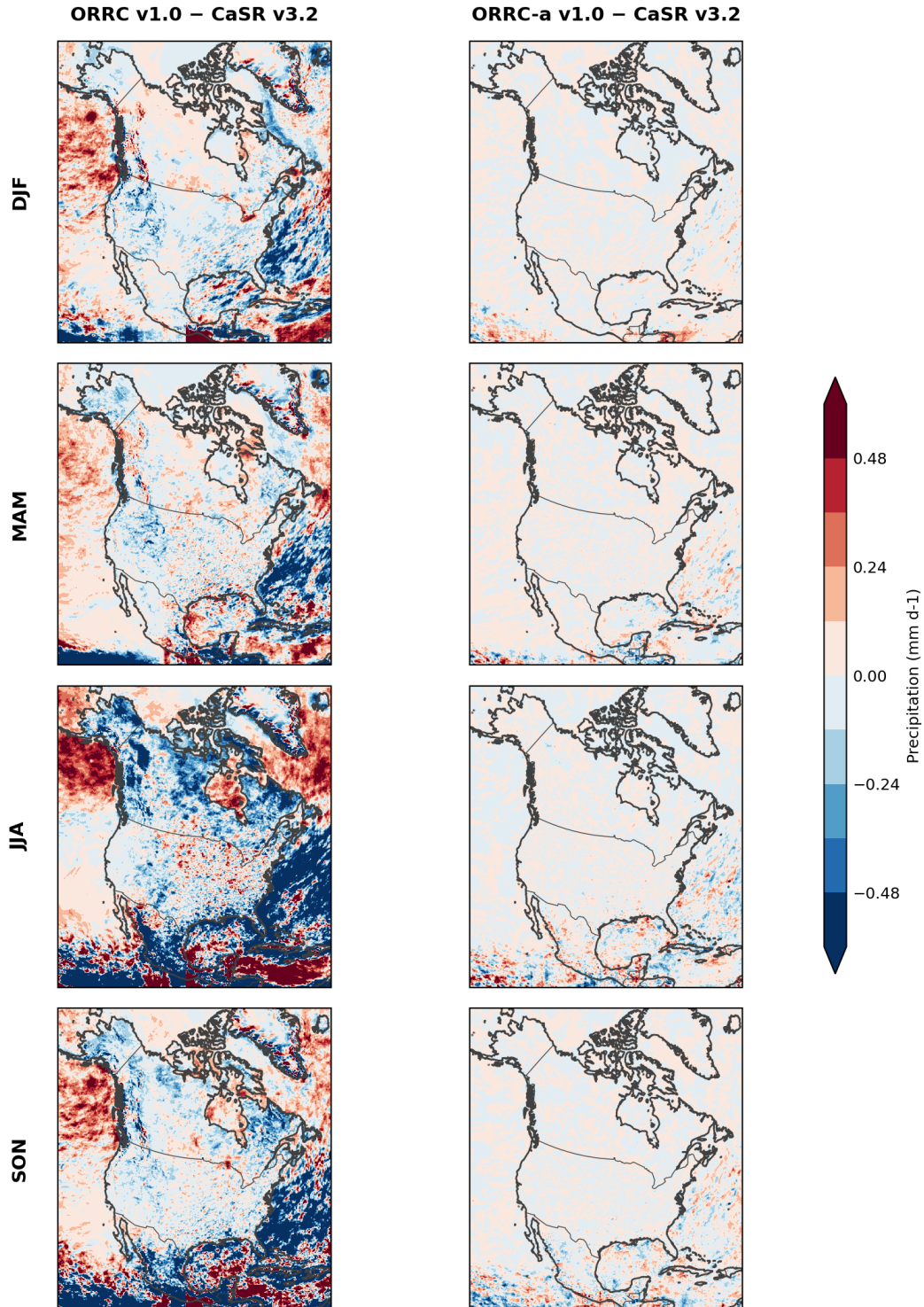


Figure 3. Seasonal mean precipitation bias relative to CaSR v3.2 over 2020–2024 for ORRC v1.0 (left column) and ORRC-a v1.0 (right column), shown for winter (DJF), spring (MAM), summer (JJA), and fall (SON) across North America. Bias is computed as the difference between the seasonal mean precipitation in each ORRC product and the corresponding seasonal mean precipitation in CaSR v3.2.

## 2.4 Evaluation against CaSR v3.2

We evaluated ORRC v1.0 and ORRC-a v1.0 against CaSR v3.2 over 2020–2024 using a consistent set of diagnostics for all variables, including regional Taylor diagrams and spatial maps of error statistics. The objective of this comparison was to assess the degree of similarity between the products, rather than to identify which product performs better.

For the regional evaluation, diagnostics were summarized over the 31 Bukovsky regions shown in Figure 4 (Bukovsky, 2011). Figure 5 illustrates this evaluation for monthly mean precipitation variability, comparing ORRC v1.0 with CaSR v3.2 across the Bukovsky regions. The diagram shows that ORRC v1.0 captures the regional pattern of precipitation variability reasonably well overall, with generally high correlations, although the normalized standard deviations are often lower than 1, indicating weaker variability than in CaSR v3.2 in many regions. Taylor diagrams for the other variables, as well as for ORRC-a v1.0, are available in the ORRC GitHub repository.

## Bukovsky Regions

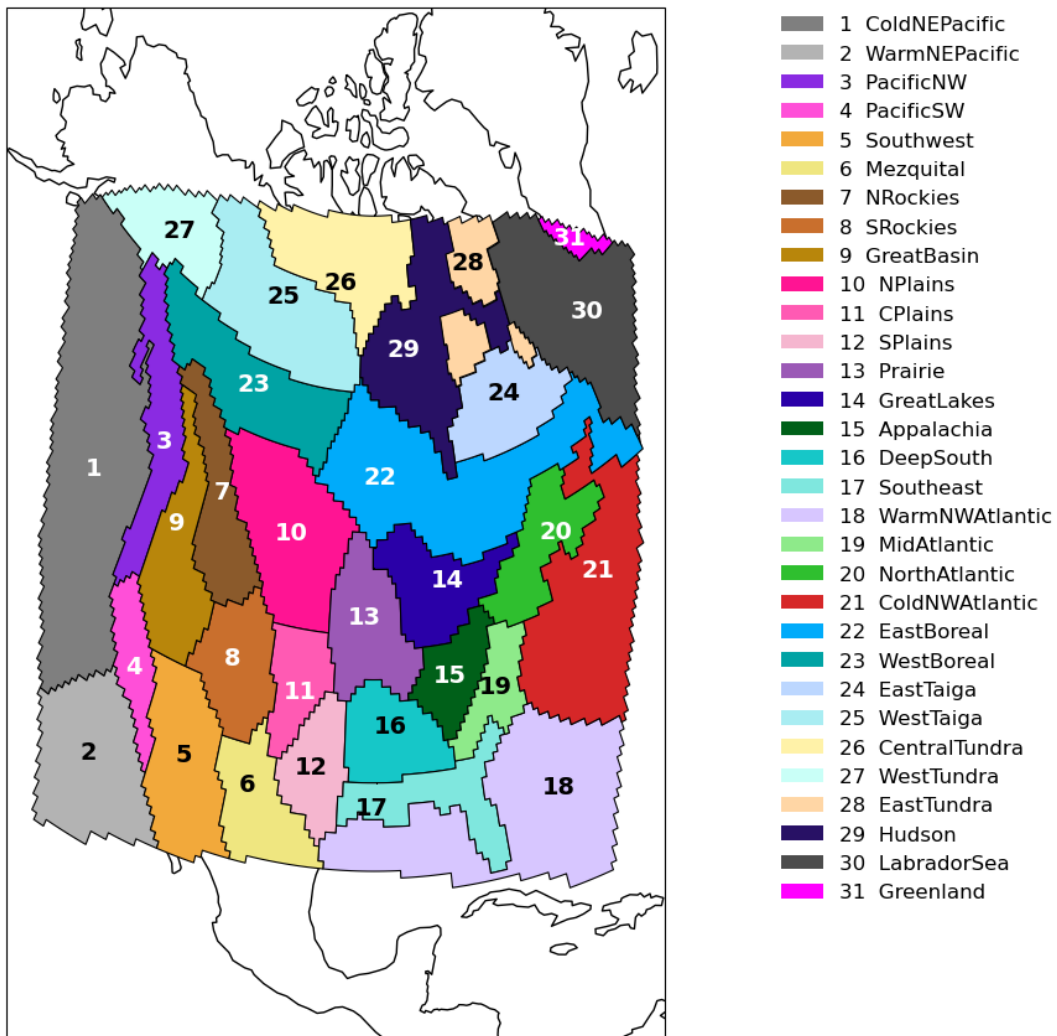


Figure 4. Bukovsky regions used for the regional evaluation of ORRC v1.0 and ORRC-a v1.0 against CaSR v3.2.

## Precipitation (2020-2024)

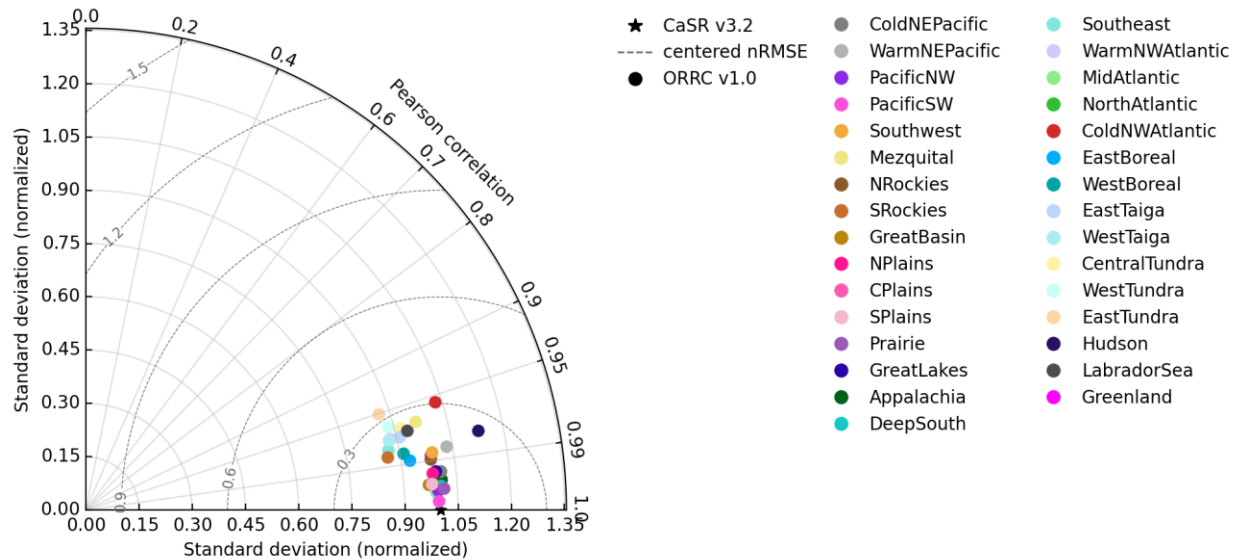


Figure 5. Taylor diagram comparing ORRC v1.0 and CaSR v3.2 monthly mean precipitation over 2020–2024 for the 31 Bukovsky regions. For each region, the statistics are computed from the time series of spatially averaged monthly precipitation. Radial distance indicates the normalized standard deviation, angle indicates the Pearson correlation coefficient and dashed contours show the centered, normalized root-mean-square error (centered nRMSE). The star marks the CaSR v3.2 reference and coloured circles denote the individual Bukovsky regions.

Figure 6 provides a full-domain spatial evaluation of ORRC v1.0 precipitation over 2020–2024 based on seasonal variability in monthly mean precipitation. The left column shows the seasonal standard deviation of monthly mean precipitation in CaSR v3.2, highlighting stronger variability along the Pacific and Atlantic coasts and in lower-latitude regions, and weaker variability over much of the continental interior and the Arctic. The normalized RMSE patterns in the right column show that ORRC v1.0 generally reproduces the broad spatial structure of this variability well, with mostly moderate errors across the domain. Larger normalized errors occur mainly in some coastal and low-variability regions, where modest RMSE values can translate into large normalized values because the reference seasonal standard deviation is small. Grid cells with very low CaSR v3.2 variability are therefore hatched in the nRMSE maps. Corresponding spatial diagnostics for the remaining variables and for ORRC-a v1.0 are available in the ORRC GitHub repository.



## Precipitation (2020-2024)

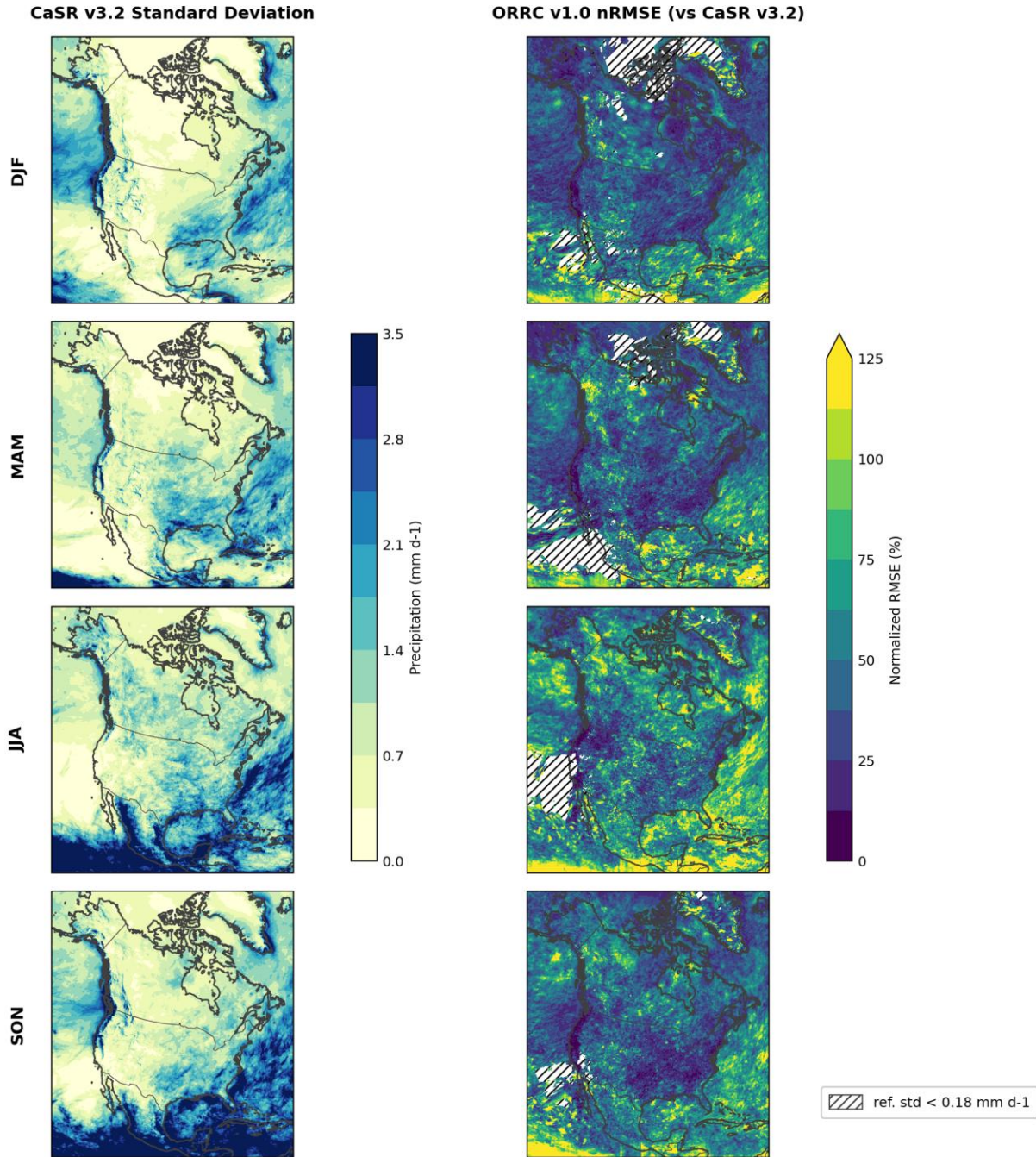


Figure 6. Seasonal standard deviation of monthly mean precipitation from CaSR v3.2 over 2020–2024 (left column) and the corresponding normalized RMSE of ORRC v1.0 relative to CaSR v3.2 (right column) for winter (DJF), spring (MAM), summer (JJA), and fall (SON) across North America. The normalized RMSE is defined as the seasonal RMSE of monthly mean precipitation divided by the seasonal standard deviation of the CaSR v3.2 reference. Grid cells where the CaSR v3.2 seasonal standard deviation is below 0.19 mm/day are hatched, indicating areas of very low reference variability where normalization can yield artificially inflated values. The 0.19 mm/day threshold corresponds to the 5<sup>th</sup> percentile of positive, finite seasonal standard deviation values of monthly mean CaSR v3.2 precipitation pooled across seasons and grid cells.

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